

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2460221.6712 +/- 0.0013 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.2108 +/- 0.0043
Transit depth (Rp/Rs)^2	4.44 +/- 0.18 [%]
Orbital Inclination (inc)	89.22 +/- 0.5
Airmass coefficient 1 (a1)	1.313 +/- 0.015
Airmass coefficient 2 (a2)	-0.253 +/- 0.011
Scatter in the residuals of the lightcurve fit is	1.13 %
Best Comparison Star	None
Optimal Aperture	4.93
Optimal Annulus	11.6
Transit Duration (day)	0.0983 +/- 0.0013

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.2108 ± 0.0043 vs 0.1592 (deviation 12.0σ)
Duration fit vs published	0.0983 d vs 0.0946 d (deviation 2.86σ)
Mid-time O–C	1.4 min
Depth SNR	49.0
Residual scatter	1.13 %

Light curve

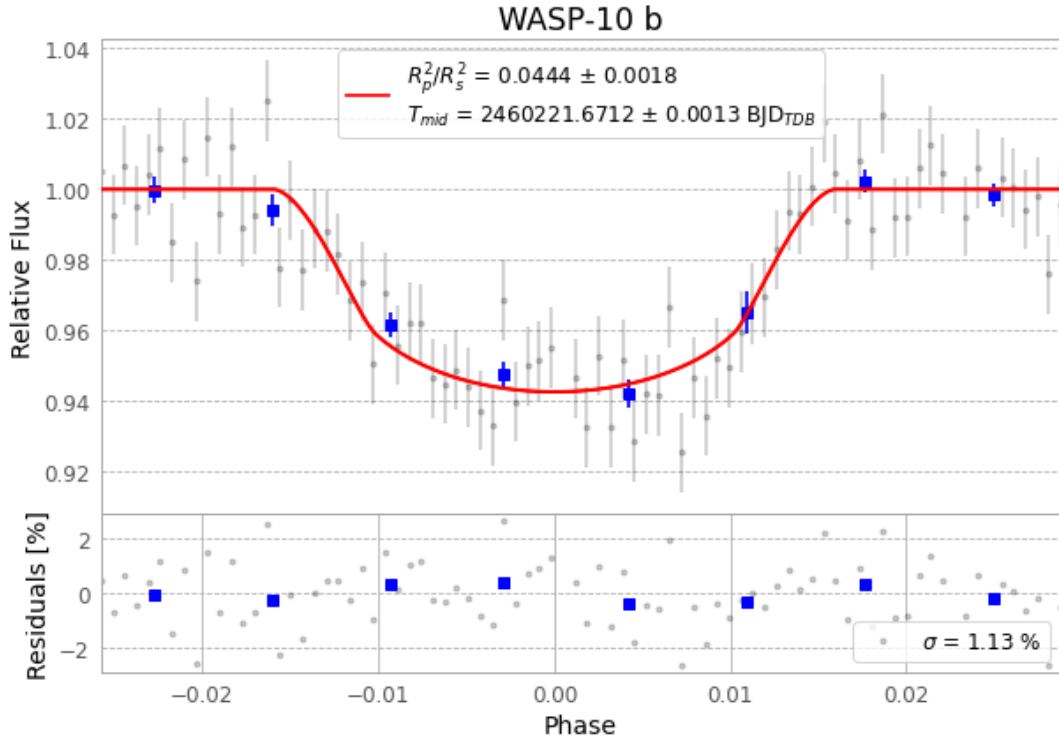


Figure 1 — FinalLightCurve_WASP-10 b_2023-10-03.png (SHA-256 2ed864698f127a8c...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [294, 234], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 d9ff1c9ddedd0795...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

82 FITS frames (54 MB), WASP-10231004020315.FITS → WASP-10231004060615.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-10-231004.log (6eb039a5af09...), FinalLightCurve_WASP-10 b_2023-10-03.png (2ed864698f12...), FinalLightCurve_WASP-10 b_2023-10-03.csv (316dd995d81e...)

Compute resources

Wall time 125.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-27 23:47Z.